KARIN KJELLGREN

Albert-Ueberle-Str. 2, 69120 Heidelberg, Germany, karin.kjellgren01@stud.uni-heidelberg.de

EDUCATION

09/2023 – Present
Ph.D. in Astrophysics
Heidelberg University
Advisors: Philipp Girichidis, Ralf Klessen
Preliminary thesis title: *The impact of cosmic rays in Milky Way-like galaxies and their gamma-ray signatures*

08/2021 – 06/2023
Master's Programme in Astronomy
Stockholm University
Advisors: Sambit Giri, Garrelt Mellema
Thesis: *Modeling the 21-cm bias from the Epoch of Reionization using Machine Learning*

04/2022 – 10/2022
Erasmus Student Exchange Programme
Heidelberg University

08/2017 – 06/2020
Bachelor's Programme in Astronomy
Stockholm University
Advisors: T. Emil Rivera-Thorsen, Angela Adamo
Thesis: *Physical properties of recombination nebulae in a gravitationally lensed, Lyman continuum emitting starburst at z=2.37 from rest-frame optical line emission*

08/2014 – 05/2017
International Baccalaureate
Kungsholmens gymnasium
- Bilingual diploma in Swedish and English

FIRST-AUTHOR PUBLICATIONS

The diffuse gamma-ray sky of a Milky Way analog: Local diversity and global constraints, **Kjellgren, K.,** Girichidis, P., Werhahn, M., et al., 2026, A&A, 710, A163

The dynamical impact of cosmic rays in the Rhea magnetohydrodynamic simulations, **Kjellgren, K.,** Girichidis, P., Göller, J., et al., 2025, A&A, 700, A124

PUBLICATIONS AS CO-AUTHOR

Introducing the Rhea simulations of Milky-Way-like galaxies I: Effect of gravitational potential on morphology and star formation*, Göller, J., Girichidis, P., Brucy, N., et al. (incl. **Kjellgren, K.**), 2025, A&A, 704, A331

The Connection Between Galactic Outflows and the Escape of Ionizing Photons, Mainali, R., Rigby, J. R., Chisholm, J., et al. (incl. **Kjellgren, K.**), 2022, ApJ, 940, 2

INVITED & CONTRIBUTED TALKS

1. *LMU Klaus Dolag's group seminar*, Oct. 22, 2026, Munich, Germany, **invited seminar**
2. *MPA Cosmology seminar*, Oct. 20, 2026, Munich, Germany, **invited seminar**
3. *Cosmic Rays 4,* Oct. 5 – 9, 2026, Pisa, Italy, **contributed talk**
4. *BUGS2026: Building Galaxies from Scratch II,* Sep. 7 – 11, 2026, Heidelberg, Germany, **contributed talk**
5. *9th Heidelberg International Symposium on High-Energy Gamma-Ray Astronomy,* Aug. 31 – Sep. 4, 2026, Heidelberg, Germany, **contributed talk**
6. *Workshop on Zoom-in Views of Galaxy Disks Across Cosmic Time: Bridging Simulations and Observations*, Jun. 1 – 19, 2026, Stockholm, Sweden, **contributed talks**
7. *Lund Observatory Division of Astrophysics weekly seminar,* Apr. 16, 2026, Lund, Sweden, **invited seminar**
8. *ECOGAL weekly seminar,* Oct. 4, 2025, online, **invited seminar**
9. *MIST2025 - Cosmic turbulence and Magnetic fields: physics of baryonic matter across time and scales,* Cargése, France, Sep. 29 – Oct. 3, 2025, **contributed talk**
10. *MPIK group seminar*, Jun. 3, 2025, Heidelberg, Germany, **invited seminar**
11. *Physics of Star Formation: ISM dynamics and star formation*, Dec. 3 – 6, 2024, Heidelberg, Germany, **contributed talk**
12. *Cosmic Rays 3*, Oct. 22 – 24, 2024, Florence, Italy, **contributed talk**
13. *8th Heidelberg International Symposium on High-Energy Gamma-Ray Astronomy*, Sep. 2 – 6, 2024, Milan, Italy, **contributed talk**

OTHER SCHOOLS & CONFERENCES

Durham Advanced School on Astrophysical Simulations, Sep. 15 – 19, 2025, Durham University, UK

Magnetic Fields and Cosmic Rays across Diverse Scales: What's Next?, Sep. 8 – 12, 2025, CfA, Boston, USA

18th Potsdam Thinkshop, Jul. 14 – 18, 2025, AIP, Germany

ALMABO 2024: View on the multi-phase interstellar medium in galaxies, Sep. 9 – 13, 2024, Bologna, Italy

ECOGAL school on "The Physics of Star Formation", Feb. 12 – 23, 2024, Les Houches, France

OUTREACH AND OTHER ACADEMIC WORK

09/2026	Member of the LOC *9th Heidelberg International Symposium on High-Energy Gamma-Ray Astronomy*
2025	Teaching assistant *Introduction to astronomy II*
04/2025	Workshop leader *Girls' Day, University of Heidelberg*
2024	Teaching assistant *Astronomical techniques*
09/2022	Volunteer *ICAS (International Council of the Aeronautical Sciences) 2022 Congress*, Stockholm
04/2018	Organizer of the "Knut Lundmark"-days *The Swedish Astronomical Youth Association*, Stockholm
09/2017	Representative at the National Museum of Science and Technology *The Swedish Astronomical Youth Association*, Stockholm

LANGUAGE SKILLS

Swedish (native), English (professional), French (B2), German (B2)